PRODUCT DATA SHEET

PU SEALANT

One Component Polyurethane Construction Sealant

DUROTECH
Engineered Construction Chemicals
Made In Australia For Over 45 Years

DESCRIPTION

PU Sealant is a novel low modulus expansion joint sealant, especially formulated to ensure bubble free cure even at very high temperature and humidity climatic conditions. The product displays excellent thixotropy allowing its use even in very large expansion joints.

It cures by reaction with atmospheric humidity to produce a joint sealant with a 50% joint movement accommodation factor and excellent adhesion on substrates traditionally problematic for PU sealants, e.g. glass, aluminum, steel, polycarbonate, etc. The extrusion rate and tooling of the sealant remain the same throughout a very wide range of temperature and humidity conditions.

COMPLIANCE WITH

- ISO-11600
- Type F - class: 25LM
- DIN-18540-F
- ASTM C920
- U.S. Federal Specification TT-S-00230C
- Type II Class A

PU Sealant is CE certified according to
EN 15651-1 (Sealants for Facades) and
15651-4 (Sealants for Floor Joints with Foot traffic).

RECOMMENDED FOR

- In-situ concrete
- Precast concrete panels
- Brick and block work
- Water tanks and swimming pools
- Metal frames
- Aluminum windows and panels
- Irrigation channels
- Glass
- Granite & marble

LIMITATIONS

- Not recommended for direct application on unsound substrates.
  In this case, the substrate must be primed with POLYPRIME, which will re-enforce the concrete and produce a strong durable substrate for sealant application.
- Highly porous substrates, dusty surfaces or poorly compacted concrete, must have their porous bond area surfaces thoroughly sealed to avoid the possibility of air bubbles being blown into the uncured sealant if the substrate temperature rises.
- PU Sealant is suitable for UV exposure and has been certified for external use according to EN 15651-1 & 4, however if colour retention and no chalking is required, it is recommended that the sealant be painted with good quality acrylic paint.

FEATURES & BENEFITS

- No bubbling/swelling upon curing in difficult climatic conditions.
- Excellent Thixotropy
- Excellent adhesion on almost any type of surface, with or without the use of special primers.
- Excellent extrusion, tooling and storage stability over wide range of climatic conditions.
- Excellent chemical resistance, suitable for sealing joints in swimming pools and chemically treated water.
- Low modulus, joint movement accommodation 50%.
- Microorganism and fungus resistant.
- Excellent heat resistance, suitable for application where exposure to temperatures >60°C take place.
- Resistance to cold: The sealant remains elastic even down to -40°C.

AUSTRALIAN MADE SINCE 1977

SINCE 1977 Trialled • Tested • Approved

APPLICATION PROCEDURE

Clean joint thoroughly, and ensure that no oil, grease and wax contaminants, silicone remains are present.

For many applications, primer is not required. In the case of application on very porous substrates, bond area surfaces thoroughly to avoid the possibility of air bubbles being blown into the uncured sealant if the substrate temperature rises. The recommended primer is POLYPRIME.

Apply backing material such as open cell polyurethane or a closed cell polyethylene backing rod. Although both types of backing rod are recommended, care must be taken when using the closed cell polyethylene rod that the outer skin not be punctured as in rising temperature conditions it may cause bubbling. Backing rod application is important as it ensures that the correct width to depth ratio is achieved to provide a firm backing against which the sealant can be tooled off.

Slide the sealant into the applicator gun, cut off the very end of the sealant packaging and fit the gun with the nozzle that has been cut to deliver the right bead size.

Extrude the sealant into the joint ensuring that no air is trapped in the joint.

Tooling is recommended immediately after the application of sealant.

The ratio width to depth should be 2:1 subject to a minimum depth of 10mm.

CONSUMPTION

Linear meters per 600cc sausage:

WIDTH / DEPTH	5mm	10mm	15mm	20mm	25mm
5mm	**24**	**12**			
10mm			**4**	**3**	**2.4**
15mm					**1.6**

CONSUMPTION

600cc sausage
300cc cartridge

SHELF LIFE

12 months minimum for 600 cc sausages and 9 months for 300 cc cartridges in the original packaging when stored in dry places and at temperatures of 5-25°C. Once opened, use as soon as possible.

TECHNICAL SPECIFICATIONS

PROPERTY	UNITS	METHOD	SPECIFICATION
Specific weight	gr/cm^3	ASTM D1475 / DIN 53217 / ISO 2811, @ 20°C	+/- 1.49
Tack free time, @ 77° F (25°C) & 55% RH	hours	-	2.5-3.5
Cure Rate	Mm/day	-	2-3
Service temperature	°C	-	-40 to 80
Hardness Modulus at 100% elongation	Shore A (N/mm^2)	ASTM D2240 / DIN 53505 / ISO R868 ASTM D412 / EN-ISO-527-3	±27 0.3
Elongation	%	ASTM D412 / EN-ISO-527-3	>700
QUV Accelerated Weathering Test(4hr UV, at 60°C (UVB- Lamps) & 4hr COND at 50°C)	-	ASTM G53	passed (after 2000hr)
Thermal Resistance (100 days, 80°C)	-	EOTA TR011	passed

PROPERTY	UNITS	METHOD	SPECIFICATION
Toxicity	-	-	No restrictions after full cure
Resilience	%	DIN 52458	>90
Hydrolysis (8% KOH, 15 days @ 50°C)	-	-	No elastomeric property change
Hydrolysis (H2O, 30 days- cycle 60-100°C)	-	-	No elastomeric property change
HCl (PH=2, 10 days @RT)	-	-	No elastomeric property change
Adhesion to concrete	kg/cm^2	ASTM D4541	>20

EU-DECLARATION OF PERFORMANCE

In accordance with Annex III of Regulation (EU) No.305/2011 (Construction Product Regulation)

For the product "HYPERFLEX"

No: CPR-5113/850/14-3

CPR-5113/850/14-4

The notified body (0761):

Materiaiorufarsai (MPA)

fur das Bauwesen BeethovenstratBe 52

D-38106 Braunschweig

EN 15651 – part 1 & part 4

The product is used as a single-component, polyurethane-based joint sealant

PROPERTY	PERFORMANCE	HARMONIZED TECHNICAL SPECIFICATION
Reaction to fire	Class E	EN ISO 11925-2
Elastic recovery (%)	>70%	EN ISO 7389
Resistance to flow (mm)	≤3mm	EN ISO 7390
Tensile properties – secant modulus- at 23°C	≤0.4MPa	EN ISO 8339
Tensile properties – secant modulus- at -30°C	≤0.9MPa	EN ISO 8339
Tensile properties at maintained extension	NF	EN 8340
Adhesion/cohesion properties at variable temperature	NF	EN ISO 9047
Loss of mass/volume	≤10%	EN ISO 10563
Tensile properties at maintained extension after immersion in water (4 days)	NF	EN ISO 10590
Tensile strength (movement capacity 50%)	NF	EN ISO 8340

Outdoor Requirements:

Tensile properties at maintained extension after	NF	EN ISO 10590
immersion in water (28 days) Tensile properties at maintained extension after	NF	EN ISO 10590
immersion in saltwater (28 days) Adhesion/cohesion properties after exposure to heat,	NF	EN ISO 11431
water and artificial light through glass		

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